NAEEM KHAN

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PERSONAL STATEMENT

MSc Automotive Engineering student at Cranfield University with three years' automotive engineering experience at Ford Motor Company and technical experience in vehicle dynamics simulation, Driver-in-the-Loop systems and performance analysis. Developed real-time MATLAB/Simulink vehicle models covering longitudinal, lateral and roll dynamics, integrated with Unreal Engine for driver simulation and verified against first-principles calculations. Combines vehicle simulation and telemetry analysis with production engineering experience in physical validation, DVP&R, DFMEA and cross-functional problem solving.

KEY ACHIEVEMENTS

- Managed design-verification sign-off for 10 wheel designs and one jack design for the Ford Transit V710 programme under production timing requirements (Ford Motor Company, 2024)
- Achieved two consecutive fourth place finishes with Atlassian Williams F1 Team Gaming in the Formula 1 Sim Racing teams' championship against nine teams, applying structured telemetry and performance analysis (2024 – 2026)

CAREER HISTORY

Wheels & Jack Design & Release Engineer: Ford Motor Company (Via Caresoft Global) | Essex | UK
(September 2022- August 2025)

Ford's UK commercial-vehicle engineering centre supporting global Transit programmes, supplier-led component development and production design release.

- Reviewed supplier 3D CAD models and 2D production drawings in CATIA V5 against fit, interface, manufacturing and GD&T requirements, identifying design issues and supporting component approval for production release.
- Defined and updated DVP&R test requirements and acceptance criteria, assessed technical risk, coordinated validation activity and reviewed test evidence, supporting design-verification sign-off for 10 wheel designs and one jack design within programme timing.
- Resolved component-development issues across design, test and supplier teams, balancing technical performance, reliability, cost and timing to enable components to progress through validation and production release.

EDUCATION

MSc Automotive Engineering: Cranfield University, Cranfield, UK (September 2025 – September 2026)

- **Modules:** Vehicle Dynamics, Automotive Control & Simulation, Powertrain Performance & Simulation, Vehicle Design Propulsion and Performance, Automotive Aerodynamics, Vehicle Electrification & Hybridisation
- **Group Design Project:** Conceptual Design of a Quadricycle Vehicle & Mobility Service: Responsible for vehicle dynamics and wheel/tyre integration, using first-principles calculations, MATLAB/Simulink, IPG CarMaker and CATIA to evaluate handling and component loads and inform wheel/tyre selection for the final vehicle concept.
- **Individual Thesis:** Heavy Goods Vehicle Driver Simulator for Roll Stability Analysis & Driver Training: Developed and verified a real-time driver-in-the-loop framework model integrated with Unreal Engine visualisation using MATLAB/Simulink to communicate the vehicle model, enabling handling and rollover-stability analysis across varying vehicle mass and centre-of-gravity conditions. Created the Cotswold Airfield environment in Blender for integration with the simulator.

EDUCATION - CONTINUED

MEng Mechanical Engineering: Lancaster University, Lancaster, UK (October 2017 – June 2021)

- **Modules:** Control Engineering, Vibration Analysis, System Design & Modelling, Fluid Dynamics, Thermodynamics, Engineering Materials, Mechanics & Actuators, Engineering Mathematics, Composites
- **Grade:** First-class Honours - 18/24 (75%)
- **Group Project:** Formula Student: - Wheel Assemblies and Structures Engineer: Designed wheel-assembly and suspension-interface components including uprights and mounting brackets using SolidWorks and first-principles load calculations; verified structural performance in ANSYS and delivered manufacture-ready drawings with tolerances and design for manufacture considerations.

VOLUNTEERING

Sim-Racing Engineer: Atlassian Williams F1 Team Gaming | Remote (January 2022- August 2026)

Official esports division of Williams Racing, competing across Formula 1 Sim Racing and iRacing endurance special events. Role focuses on simulator-based performance preparation through telemetry analysis, structured driver feedback, setup evaluation and race event support.

- Analysed performance across 20+ circuits using MoTeC and bespoke telemetry tools to identify lap-time losses and prioritise setup changes for simulator and race preparation.
- Assessed vehicle balance, rotation, braking and traction using telemetry and structured driver feedback, translating findings into setup recommendations including spring and anti-roll-bar stiffness changes.
- Supported simulator-based setup development and race preparation by translating driver feedback into engineering changes, contributing to consecutive fourth-place finishes in the Formula 1 Sim Racing Teams' Championship against nine teams.

SKILLS, INTERESTS & EXTRACURRICULAR ACTIVITIES

- **Languages:** Fluent English
- **Engineering Processes:** DVP&R, DFMEA, GD&T, Test planning, design verification and supplier coordination
- **Software:** MATLAB/Simulink, IPG CarMaker, MoTeC, Unreal Engine, Python, CATIA V5, SolidWorks, FreeCAD, ANSYS Structural, OptiStruct, Teamcenter, Blender, Microsoft Project and Microsoft Office.
- **Memberships:** Associate Member, Institution of Mechanical Engineers (IMechE)
- **Individual Interests:** Sim racing, vehicle simulation, motorsport engineering, Formula 1, endurance racing, karting, table tennis and board games.